Complete the following instructions. Execute each one and write a report on the answers to the instructions. First complete the following tasks to create the database and the tables.

1. Create a database called “School”. Write the corresponding query for this in your report.
2. Create a table titled “Students” with the following constraints. Write the corresponding query for this in your report.

Field	Type	Constraints
student_id	int	Primary key, not null, auto_increment
student_name	varchar(50)	Not null
class	Varchar(20)	Not null
date_of_birth	date	Not null
gender	Varchar(10)	Not null

3. Create a table titles “Teachers” following constraints. Write the corresponding query for this in your report.

Field	type	constraints
teacher_id	int	Primary key, not null, auto_increment
teacher_name	Varchar(50)	Not null
subject	Varchar(50)	Not null
hire_date	date	Not null

4. Create a table titled “Results” with the following constraints. Write the corresponding query for this in your report.

field	type	constraints
result_id	int	Primary key, not null, auto_increment
student_id	int	Not null
subject	Varchar(50)	Not null
Marks_obtained	int	Not null default 0
Total_marks	int	Not null default 100

5. Populate the table “Students” with the following data. Write the corresponding insertion query for this in your report.

student_name	class	date_of_birth	gender
Arif Hasan	10	2007-02-15	Male

Nusrat Jahan	9	2008-09-05	Female
Tamim Rahman	10	2007-06-20	Male
Farzana Akter	8	2009-03-30	Female
Rafiul Islam	9	2008-12-11	Male

6. Populate the table "Teachers" with the following data. Write the corresponding insertion query for this in your report.

teacher_name	subject	hire_date
Khalid Hossain	Math	2015-01-10
Rina Akter	English	2017-08-20
Ahsan Habib	Science	2016-03-15
Sumaiya Haque	History	2018-11-05

7. Populate the table "Results" with the following data. Write the corresponding insertion query for this in your report.

student_id	subject	marks_obtained	total_marks
1	Math	88	100
2	Math	92	100
3	English	75	100
4	Science	60	100
5	History	85	100
1	Science	78	100
2	English	89	100

After creating and populating the tables, write queries in your report to do the following tasks.

a) **Query 1: Display all students.**
Sample Output:

student_id	student_name	class	date_of_birth	gender
1	Arif Hasan	10	2007-02-15	Male
2	Nusrat Jahan	9	2008-09-05	Female
3	Tamim Rahman	10	2007-06-20	Male
4	Farzana Akter	8	2009-03-30	Female
5	Rafiul Islam	9	2008-12-11	Male

b) **Query 2: List all teachers and their subjects.**

Sample Output:

teacher_name	subject
Khalid Hossain	Math
Rina Akter	English
Ahsan Habib	Science
Sumaiya Haque	History

c) **Query 3: Find students born after 2008.**

Sample Output:

student_name	class	date_of_birth
Nusrat Jahan	9	2008-09-05
Farzana Akter	8	2009-03-30
Rafiul Islam	9	2008-12-11

d) **Query 4: Show results where marks obtained > 80.**

Sample Output:

student_id	subject	marks_obtained
1	Math	88
2	Math	92
5	History	85
2	English	89

e) **Query 5: Count number of students in each class.**

Sample Output:

class	student_count
10	2
8	1
9	2

f) **Query 6: List students with their total marks across all subjects.**

Sample Output:

student_name	total_marks
Arif Hasan	166
Farzana Akter	60
Nusrat Jahan	181
Rafiul Islam	85
Tamim Rahman	75

g) **Query 7: Find students who scored more than 90 in any subject.**

Sample Output:

student_name	marks_obtained
Nusrat Jahan	92

h) **Query 8: Find average marks obtained per student.**

Sample Output:

student_name	avg_marks
Arif Hasan	83.0000
Farzana Akter	60.0000
Nusrat Jahan	90.5000
Rafiul Islam	85.0000
Tamim Rahman	75.0000

i) **Query 9: Find the highest scoring student in total.**

Sample Output:

student_name	total_marks
Nusrat Jahan	181

j) **Query 10: Subject-wise average marks and highest mark.**

Sample Output:

subject	avg_marks	highest_mark
English	82.0000	89
History	85.0000	85
Math	90.0000	92
Science	69.0000	78

SOLUTION

1. Create the database

```
CREATE DATABASE SchoolDB;
```

2. Create Students table

```
CREATE TABLE Students (
   student_id INT PRIMARY KEY NOT NULL AUTO_INCREMENT,
   student_name VARCHAR(100) NOT NULL,
   class VARCHAR(20) NOT NULL,
   date_of_birth DATE NOT NULL,
   gender VARCHAR(10) NOT NULL
);
```

3. Create Teachers table

```
CREATE TABLE Teachers (
   teacher_id INT PRIMARY KEY NOT NULL AUTO_INCREMENT,
   teacher_name VARCHAR(100) NOT NULL,
   subject VARCHAR(50) NOT NULL,
```

```
    hire_date DATE NOT NULL
);
```

4. Create Results table

```
CREATE TABLE Results (
    result_id INT PRIMARY KEY NOT NULL AUTO_INCREMENT,
    student_id INT NOT NULL,
    subject VARCHAR(50) NOT NULL,
    marks_obtained INT NOT NULL DEFAULT 0,
    total_marks INT NOT NULL DEFAULT 100,
    FOREIGN KEY (student_id) REFERENCES Students(student_id)
);
```

5. Insert Students

```
INSERT INTO Students (student_name, class, date_of_birth, gender) VALUES
('Arif Hasan', '10', '2007-02-15', 'Male'),
('Nusrat Jahan', '9', '2008-09-05', 'Female'),
('Tamim Rahman', '10', '2007-06-20', 'Male'),
('Farzana Akter', '8', '2009-03-30', 'Female'),
('Rafiul Islam', '9', '2008-12-11', 'Male');
```

6. Insert Teachers

```
INSERT INTO Teachers (teacher_name, subject, hire_date) VALUES
('Khalid Hossain', 'Math', '2015-01-10'),
('Rina Akter', 'English', '2017-08-20'),
('Ahsan Habib', 'Science', '2016-03-15'),
('Sumaiya Haque', 'History', '2018-11-05');
```

7. Insert Results

```
INSERT INTO Results (student_id, subject, marks_obtained, total_marks) VALUES
```

(1, 'Math', 88, 100),

(2, 'Math', 92, 100),

(3, 'English', 75, 100),

(4, 'Science', 60, 100),

(5, 'History', 85, 100),

(1, 'Science', 78, 100),

(2, 'English', 89, 100);

Query :

a) Query 1: Display all students

SELECT * FROM Students;

student_id	student_name	class	date_of_birth	gender
1	Arif Hasan	10	2007-02-15	Male
2	Nusrat Jahan	9	2008-09-05	Female
3	Tamim Rahman	10	2007-06-20	Male
4	Farzana Akter	8	2009-03-30	Female
5	Rafiul Islam	9	2008-12-11	Male

b) Query 2: List all teachers and their subjects

SELECT teacher_name, subject FROM Teachers;

teacher_name	subject
Khalid Hossain	Math
Rina Akter	English
Ahsan Habib	Science
Sumaiya Haque	History

c) Query 3: Find students born after 2008

SELECT student_name, class, date_of_birth

FROM Students

WHERE date_of_birth > '2008-01-01';

student_name	class	date_of_birth
Nusrat Jahan	9	2008-09-05
Farzana Akter	8	2009-03-30
Rafiul Islam	9	2008-12-11

d) Query 4: Show results where marks obtained > 80

SELECT student_id, subject, marks_obtained

FROM Results

WHERE marks_obtained > 80;

student_id	subject	marks_obtained
1	Math	88
2	Math	92
5	History	85
2	English	89

e) Query 5: Count number of students in each class

SELECT class, COUNT(*) AS student_count

FROM Students

GROUP BY class;

class	student_count
10	2
8	1
9	2

f) Query 6: List students with their total marks across all subjects

SELECT S.student_name, SUM(R.marks_obtained) AS total_marks

FROM Students S

JOIN Results R ON S.student_id = R.student_id

GROUP BY S.student_name;

student_name	total_marks
Arif Hasan	166
Farzana Akter	60
Nusrat Jahan	181
Rafiul Islam	85
Tamim Rahman	75

g) Query 7: Find students who scored more than 90 in any subject

SELECT S.student_name, R.marks_obtained

FROM Students S

JOIN Results R ON S.student_id = R.student_id

WHERE R.marks_obtained > 90;

student_name	marks_obtained
Nusrat Jahan	92

h) Query 8: Find average marks obtained per student

SELECT S.student_name, AVG(R.marks_obtained) AS avg_marks

FROM Students S

JOIN Results R ON S.student_id = R.student_id

GROUP BY S.student_name;

student_name	avg_marks
Arif Hasan	83.0000
Farzana Akter	60.0000
Nusrat Jahan	90.5000
Rafiul Islam	85.0000
Tamim Rahman	75.0000

i) Query 9: Find the highest scoring student in total

```
SELECT student_name, total_marks
FROM (
   SELECT S.student_name, SUM(R.marks_obtained) AS total_marks
   FROM Students S
   JOIN Results R ON S.student_id = R.student_id
   GROUP BY S.student_name
) AS totals
ORDER BY total_marks DESC
LIMIT 1;
```

student_name	total_marks
Nusrat Jahan	181

j) Query 10: Subject-wise average marks and highest mark

```
SELECT subject,
     AVG(marks_obtained) AS avg_marks,
     MAX(marks_obtained) AS highest_mark
FROM Results
GROUP BY subject;
```

subject	avg_marks	highest_mark
English	82.0000	89
History	85.0000	85
Math	90.0000	92
Science	69.0000	78